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Problem Statement-3

Overview

Monitoring Air Pollution from Space using Satellite Observations, Ground-Based Measurements, Reanalysis Data, and AI/ML Techniques

Air pollution is a major environmental concern in many urban centers. Regular, consistent monitoring of air pollutant concentrations remains a challenge. Satellite remote sensing enables observations over broad spatial areas, allowing indirect estimation of pollutant levels. This problem statement focuses on estimating surface-level particulate matter (PM) concentrations by leveraging satellite-derived data and AI/ML-based predictive modeling approaches.

Objective:

- Estimate surface-level **Particulate Matter (PM)** concentration using **Aerosol Optical Depth (AOD)** measurements obtained from INSAT-3D/3DR/3DS satellite observations.

Expected Outcomes:

- High-resolution spatial map depicting PM concentration across India, derived from AI/ML-based estimation pipelines.

Dataset Required:

- **AOD Data:** INSAT-3D, INSAT-3DR, INSAT-3DS satellite observations.

- **Ground-Based PM Data:** Central Pollution Control Board (CPCB) station-level data.
- **Atmospheric Reanalysis Data:** Meteorological variables from sources like MERRA-2.

Suggested Tools/Technologies:

Programming tools such as **Python** or **Matlab** for data pre-processing and integration. AI/ML methods such as **Random Forest** for regression modeling and prediction.

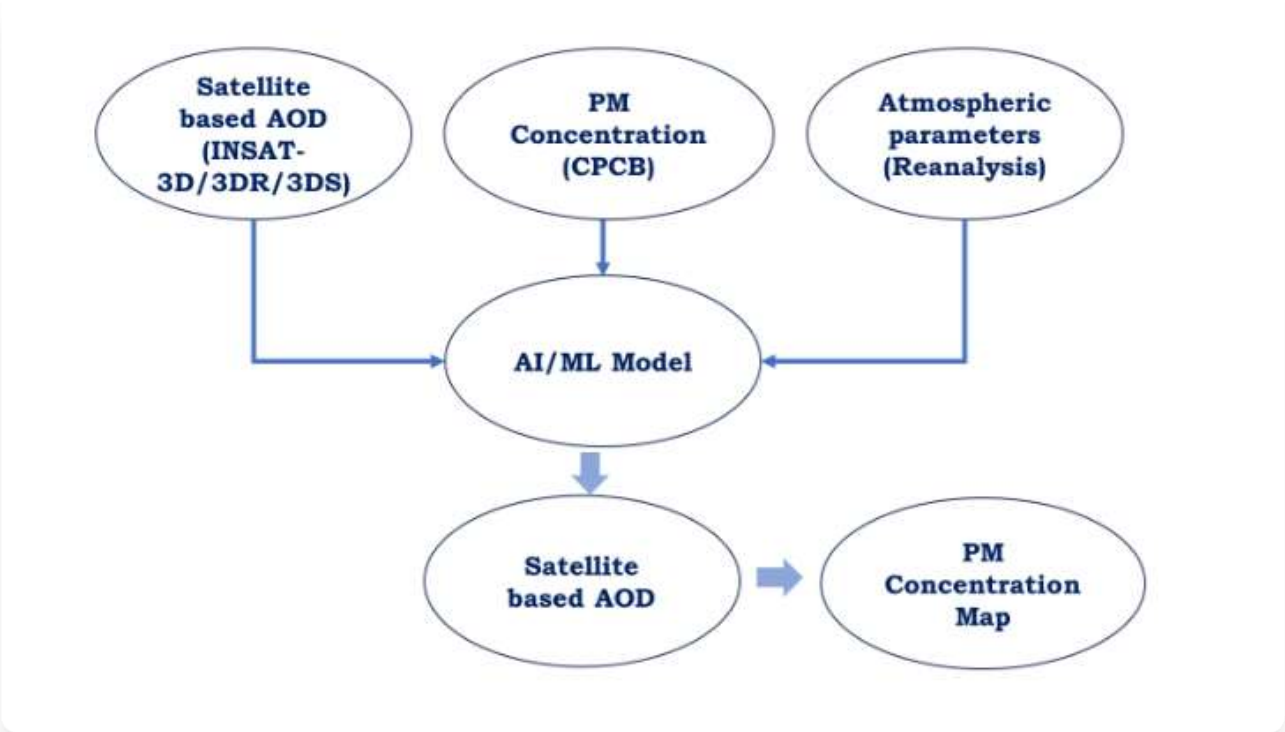
Expected Solution / Steps to be followed to achieve the objectives:

1. Generate AOD spatial maps from INSAT satellite datasets.
2. Build surface-level PM concentration data using ground-based sources like CPCB.
3. Integrate atmospheric variables from reanalysis datasets (e.g., MERRA-2).
4. Train an ML model (e.g., Random Forest) using the combined satellite, ground, and reanalysis data.
5. Predict surface PM levels from AOD using the trained model.
6. Validate the predictions with ground truth from CPCB measurements.

Evaluation Parameters:

Accuracy of the estimated PM values must be assessed by comparing them with trusted ground-based measurements from CPCB stations.

Image Representing Problem Statement:


Satellite based AOD (INSAT-3D/3DR/3DS)
PM Concentration (CPCB)
Atmospheric parameters (Reanalysis)
AI/ML Model
Satellite based AOD
PM Concentration Map